

## Beer from the early dynasties (3500–3400 cal B.C.) of Upper Egypt, detected by archaeochemical methods

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**Abstract.** Physical and chemical analyses of beer residues recovered from a vat site at Hierakonpolis (Upper Egypt) were carried out. Radiocarbon dates of the residues suggest a dating of 3500–3400 cal B.C. and are believed to represent the oldest known beer in the world. Macroscopic and microscopic examinations of the residues revealed the presence of intact remains of grains and spikelets of wheat and barley, as well as fragments of dates and grape pips. Chemical analyses included percentages of sample ingredients, pH and total soluble ions, quantitative determinations of sugars, carboxylic acids and free amino acids. A total of 25 compounds were identified, which are components of fermentation processes that are believed to have formed in connection with the preparation of what is called Nekhen-Hoffman beer.

**Key words:** Archaeochemistry – Beer preparation – Upper Egypt – Hierakonpolis – 3500–3400 B.C.

### Introduction

According to Erman (1894, p 196), beer has for a long time been cited as ancient Egypt's national drink. It was a staple food item and a widely used metaphor for the good life for the ancient Egyptians.

Past excavations have revealed numerous tomb depictions, models, and statuettes of brewing and brewers. Among the earliest written Egyptian records of beer is *hnqt-ndmt*, "sweet beer", on the offering stela in the Dynasty III tomb of Sekherkhabau at Saqqara (Murray 1905, p 34). Kaplony (1963) also noted the mention of beer in some Early Dynastic offering contexts. Evidence of beer use extending back to the Predynastic has long been known, for example the beer from Predynastic jars at Abadiyeh (Petrie 1901, p 32).

Geller (1992, pp 19–24) outlined some aspects of the production, consumption and symbolism of beer in Egypt. He described the oldest known breweries of Egypt

and related brewing to the political economy that probably characterized the later Predynastic period.

According to Geller (1992), "Bouza" is truly an antique beer and is variously described as a thick, pale yellow beverage with a yeasty or alcoholic odour and agreeable taste. It is prepared from loaves made of wheat, barley, millet or sorghum. The loaves are broken up, soaked in water and left for fermentation to increase the alcoholic content for up to several days. Hops were not added to "Bouza", so it lacks the bitterness familiar to drinkers of European style beers and ales, but a variety of fruits or even greens may be added for flavouring, preservation, etc. The caloric and nutritive value of "Bouza" is remarkable, and it can help to correct nutritional deficiencies.

Geller (1989) conducted excavations at Hierakonpolis, which lies on the western bank of the Nile between the modern towns of Esna and Edfu (Fig. 1). In locality HK24A, a hardened black residue was recovered from vats in which intact or fragmentary cereal grains were embedded. Chemical analysis carried out by Yordan Popov (consulting chemist for the Ferrosilicon factory in Edfu, Upper Egypt) suggested that the product could have been beer.

According to Geller (1992, p 23), the radiocarbon date of the vat site (HK24A) at Hierakonpolis is 4719±34 uncal B.P., which when calibrated (according to Stuiver and Reimer, 1986), suggests a date of 3500–3400 cal B.C., transitional Amratian-Gerzean (Naqada Ib–IIa). Geller believes that the Hierakonpolis breweries are the oldest known in the world.

On the 4th of March 1989, the late Michael Allen Hoffman visited the archaeobotany laboratory of Cairo University Herbarium, for the last time. He was accompanied by Jeremy Geller who delivered us two samples (here referred to as 3H and 4H) from the vat residues recovered from site HK24A during the early 1989 season. Since then, both samples recovered from two different vats have been subjected to intensive chemical analysis, macro- and microscopic examinations.

In this account, we are reporting the results of our initial investigations carried out in the archaeobotany laboratory of Cairo University Herbarium, as well as in the

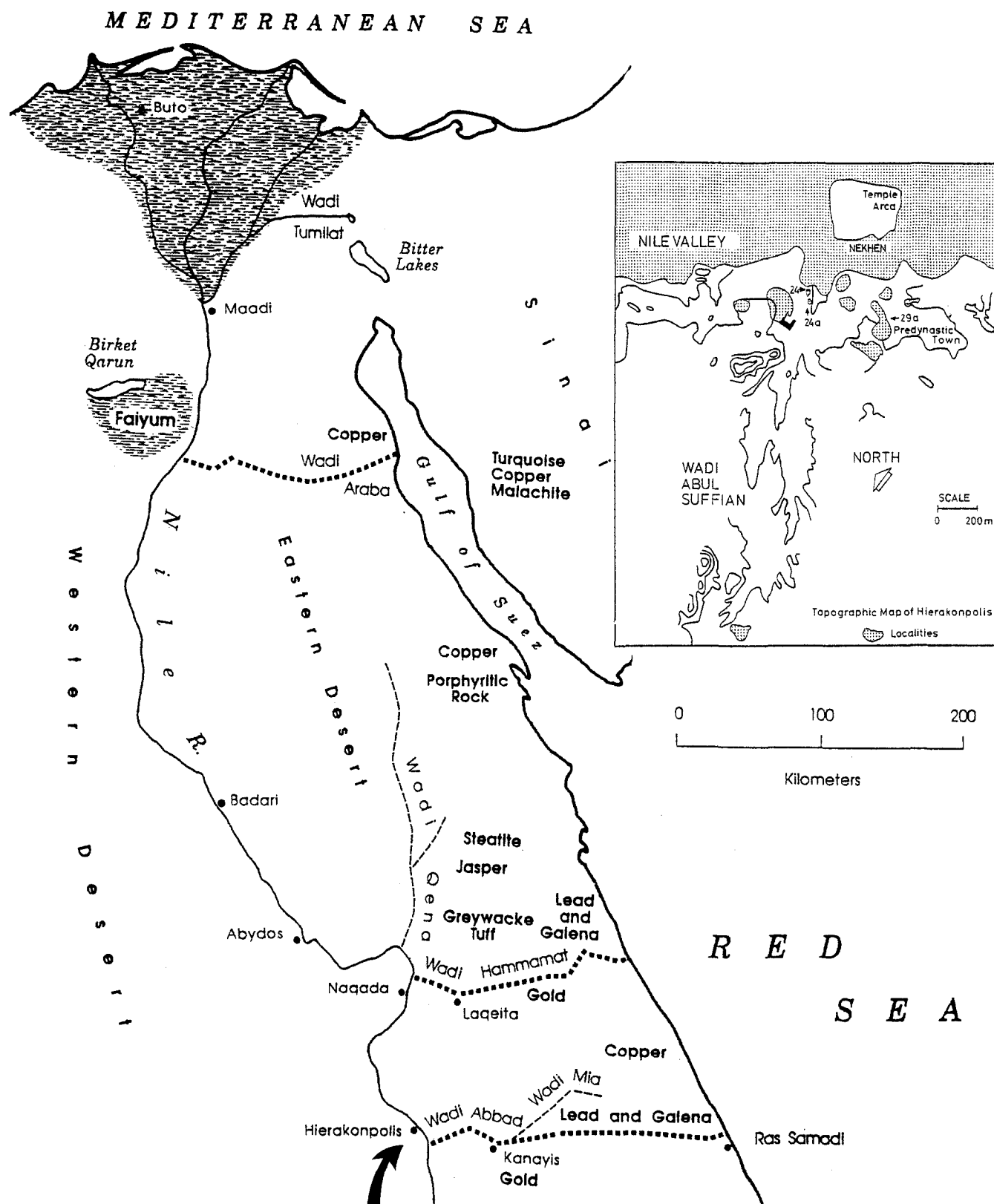


Fig. 1. Map of the study area

laboratories of the Faculty of Science, Cairo University. Further investigations were carried out by Mrs WM Amer during the spring of 1992 in the laboratories of the University of East London under the supervision of Dr John Evans. The results of these investigations will be the subject of a future publication.

The results of the present investigations enable us to confirm that the vat residue samples 3H and 4H were fermentation products of cereal grains and fruit material. The fermentation products were of a similar character to those known from the local beer "Bouza" which has been known in Egypt since early history.

We are dedicating this finding to the memory of M.A. Hoffman and calling this beverage as the "Nekhen-Hoffman Beer". Nekhen is the Archaic and Old Kingdom town where the Hierakonpolis site is situated.

## Methods

The total soluble ions and pH were measured using a Ysi model 25 conductance meter and Orion SA 520 pH meter.

The ingredients of the samples were determined following the methods of the Association of Official Analytical Chemists (AOAC 1984).

The separation of sugars, amino and organic acids was carried out according to Stumpf and Burris (1979).

The concentrated water extracts of the samples were applied to the top of a Dowex 50-X8 (H<sup>+</sup>) column. Sugars and organic acids were washed with water, while amino acids were bound and eluted with 2N HCl. Organic acids were separated from sugars by binding on the Dowex 1- X8(Cl<sup>-</sup>) column and eluting with 6M formic acid.

The procedure for O-Silylation of sugars (conversion to volatile derivatives) was carried out using a combination of TMCS and HMDS techniques as described by Biermann and McGinnis (1990). Gas chromatography (GC) was performed using a Hewlett Packard model 3392 equipped with injector, Hewlett Packard model 5890 detector and OV-17 (1.5 m x 0.4 mm) glass column. The oven temperature was programmed in a range of 140-250°C, with a rate of 5°C/min; the injector and detector temperatures were 250°C and 300°C respectively, the flow rate N=27 ml/min, H=30 ml/min, and air 300 ml/min (Bierman and McGinnis 1990).

Carboxylic acids were converted to methyl ether by ethereal diazomethane before injection to gas liquid chromatography (GLC), according to Farag and Taha (1991). GLC was performed using a GCV Pye Unicam instrument equipped with a dual flame ionization detector, coiled glass column (1.5 m x 0.4 mm) packed with acid-alkali washed and silanized Diatomite C (100-200 mesh) and coated with 10% PEGA and 10% DEGS. The oven temperature was isothermal at 150°C. The detector and injector temperatures were 220°C and 300°C respectively, flow rates N=30 ml/min, H=44 ml/min, and air=330 ml/min.

Amino acids were determined using a LKB Alpha Plus amino acid analyzer. In this high performance system, a high pressure stainless steel column with peltier heating cooling system was used, the resin was Ultrospek 8 cation exchange sodium form particle size 8 m with a bed length 200 mm and diameter of 4.6 mm. Three 0.2 N citrate buffers at pH 3.2, 4.25, and 6.45 were used for fractionation of the amino acids in a specified program that lasted for 51 minutes. The buffer flow rate was 35ml/min, while that for ninhydrin was 25 ml/minute. The reaction coil temperature was kept at 135°C, the

pressure was 15 bar for the buffer and for the ninhydrin, and the retention times and area of each peak were recognized using a Hewlett Packard 3390 integrator. Concentrations of each amino acid 1 gm/16 gm N were calculated.

## Results and Discussion

Macroscopical as well as microscopical examination revealed the presence of intact remains of grains and spikelets of wheat (*Triticum dicoccum*) and barley (*Hordeum vulgare*) as well as fragments of fruits of dates (*Phoenix dactylifera*) and pips of grapes (*Vitis vinifera*).

Figure 2 presents the possible interconversion pathways of the fermentation processes which are believed to have taken place in connection with the preparation of the Nekhen-Hoffman beer. Dotted lines designate the interactions of microorganisms; compounds identified through these investigations are underlined.

The results of physical and chemical analyses will be briefly outlined as follows:

### Sample ingredients:

The amorphous charred beer residues were subjected to intensive chemical analysis, and the percentages of the major ingredients in the samples 3H & 4H are shown in Table 1.

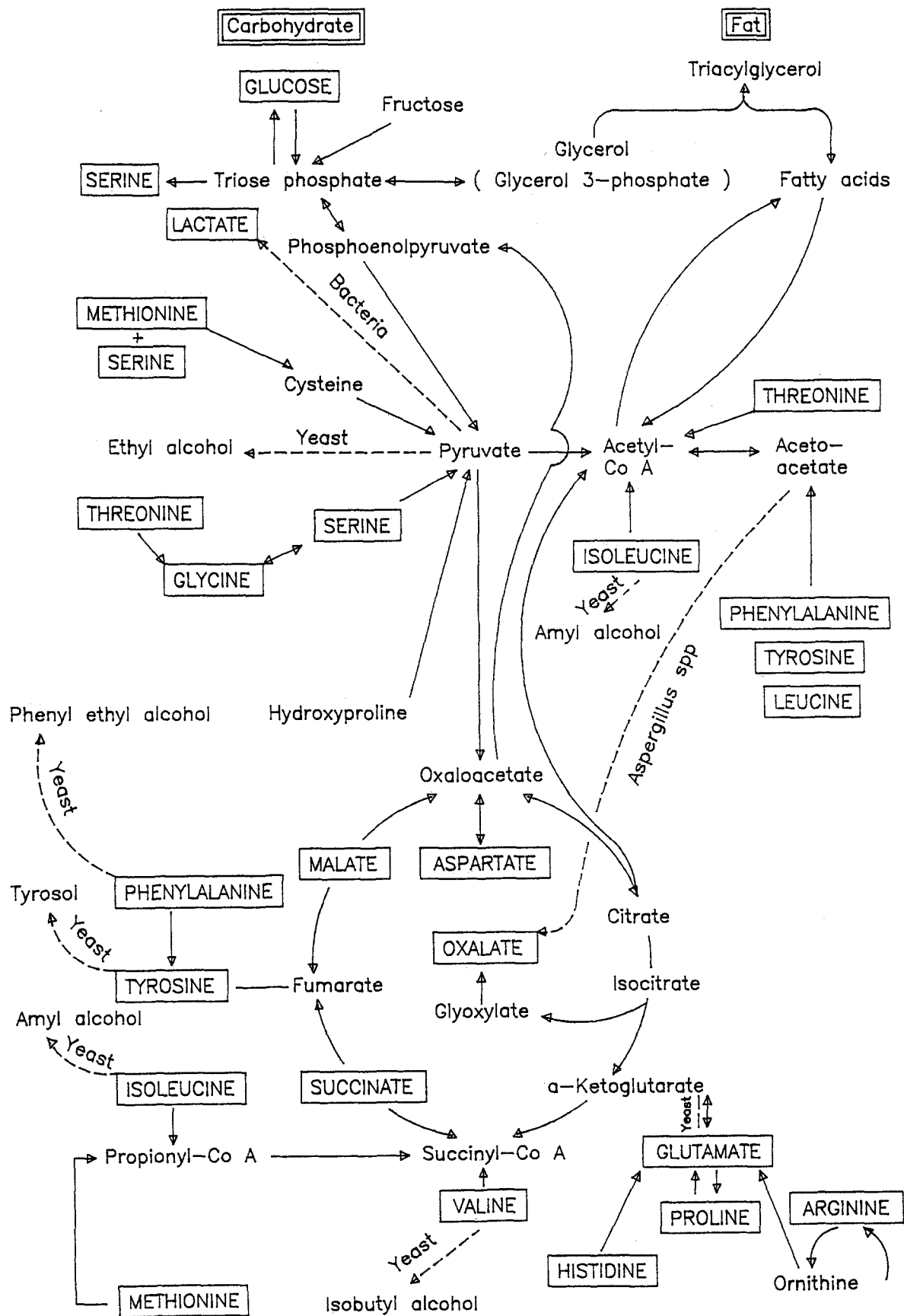
Table 1. Percentages of sample ingredients

| Ingredient             | Percentage |       |
|------------------------|------------|-------|
|                        | 3H         | 4H    |
| Ash                    | 69.22      | 72.62 |
| Carbohydrate           | 0.03       | 0.024 |
| Fibre                  | 14.93      | 12.52 |
| Lipids                 | 4.38       | 4.51  |
| Moisture               | 8.0        | 7.5   |
| Total soluble nitrogen | 2.75       | 3.07  |

It will be noticed that the constituents of both samples are closely similar. The high percentage of ash and fibre and the apparent low percentage of carbohydrates can be attributed to the fermentation process which had taken place. During this process, the decrease in carbohydrate content is due to the breakdown of sugars by the zymases of fermenting micro-organisms (such as *Saccharomyces sp.*), leading to the production of "Bouza", an idea which is proposed and supported by Morcos et al. (1973).

### pH and total soluble ions

The pH of the water extract of samples 3H and 4H was measured as 4.78 and 4.72 respectively. The acidity of both samples may be attributed to the proteolytic action



**Fig. 2.** Interconversion pathways of the major plant compounds. The identified compounds are underlined, (----)microorganisms interactions.

of moulds (Novellie 1966). Morcos et al. (1973) noticed a similar decrease in the pH of the prepared "Bouza" as fermentation progressed.

The total soluble ions as measured by conductivity meter gave the values of  $46.2 \times 10^2$  and  $47.5 \times 10^2$  for samples 3H and 4H respectively. These high values of soluble ions are an indication that complex organic compounds such as starch, cellulose, proteins and lipids were decomposed through microbial fermentation leading to the formation of simple organic and inorganic compounds.

### Quantitative determination of sugars

Table 2 shows an apparent accumulation of glucose and galacturonic acid in samples 3H and 4H as well as glucuronic acid in sample 4H.

Table 2. GC separation of silylated sugars

| Sugar and derivatives | Retention time (*R <sub>t</sub> ) | Percentage |       |
|-----------------------|-----------------------------------|------------|-------|
|                       |                                   | 3H         | 4H    |
| Unknown               | 6.75                              | 14.13      | ----  |
| Xylose                | 4.75                              | 4.85       | 6.00  |
| Mannose               | 16.22                             | 7.22       | ----  |
| Galactose             | 17.85                             | ----       | ----  |
| Glucose               | 18.70                             | 24.48      | 2.08  |
| Galacturonic acid     | 20.50                             | 22.66      | 3.81  |
| Glucuronic acid       | 21.50                             | 8.51       | 55.26 |
| Unknown               | 23.50                             | 0.70       | 14.95 |
| Unknown               | 25.70                             | 13.40      | 17.88 |

\* Samples gave the same R<sub>t</sub> as standard  $\pm 0.2\%$

Xylose and D-glucuronic acid are the decomposition products of cereal hemicellulose (Aspinall and Ross 1963). Mannose is the degradation product of the endosperm and pectic material of date seeds (Swain 1966). It is also the main sugar of Ascomycetes and accumulates after degradation of yeast cells (Harborne 1973). On the other hand, galactose and galacturonic acid may be produced after hydrolysis of pectic substances in grapes, dates and other plants by the yeast enzyme polygalacturonase (Rose and Harrison 1969).

### Quantitative determination of carboxylic acids

Table 3 shows that oxalic acid was a major constituent in both samples, representing about 50% of the total carboxylic acids identified. Considerable amounts of malic and succinic acids were determined in both samples, while tartaric and lactic acids were identified in 3H and 4H respectively.

The high oxalate level may be attributed to the fact that it is the main carboxylic acid constituent of grapes and cereals (Kasidas and Rose 1980). Oxalic acid is syn-

thesized from glyoxylic acid, which is either derived from isocitric acid (Fig. 2) or, in plants, from glycolic acid formed in photosynthesis (Lucker 1984). The accumulation of oxalic, malic and lactic acids are believed to be fermentation products of microorganisms such as *Aspergillus niger* (Martin and Steel 1955) or heterofermentive bacteria (Samuel and Cecil 1959). These organisms probably occurred as contaminants in dates and grapes. It is also possible that succinic and lactic acids are transformation products formed by the yeast *Saccharomyces sp.* from the amino acid reserves in the fruits and grains (Rose and Harrison 1969).

Table 3. GLC separation of carboxylic acids

| Carboxylic acid | Retention time (*R <sub>t</sub> ) | Percentage |       |
|-----------------|-----------------------------------|------------|-------|
|                 |                                   | 3H         | 4H    |
| Oxalic          | 1.60                              | 51.11      | 49.02 |
| Succinic        | 2.75                              | 6.20       | 24.10 |
| Tartaric        | 5.37                              | 7.01       | ----  |
| Lactic          | 7.02                              | ----       | 2.60  |
| Malic           | 12.33                             | 35.68      | 24.31 |

\* Samples gave the same R<sub>t</sub> as standard  $\pm 0.2\%$ .

### Quantitative determination of free amino acids

Table 4 shows higher accumulation of aspartic acid, threonine and glycine as well as glutamic acids in both of the samples. The percentage of the aromatic amino acid histidine in sample 3H is about double that of 4H. Some

Table 4. Free amino acids detected in beer residues

| Amino acid, NH <sub>4</sub> <sup>+</sup> | Retention time (R <sub>t</sub> ) | Percentage |      |
|------------------------------------------|----------------------------------|------------|------|
|                                          |                                  | 3H         | 4H   |
| Aspartic                                 | 6.36                             | 12.12      | 10.9 |
| Threonine                                | 7.79                             | 14.8       | 7.2  |
| Serine                                   | 8.25                             | 7.4        | 2.3  |
| Glutamic                                 | 8.86                             | 8.2        | 9.4  |
| Proline                                  | 10.36                            | 0.1        | 0.2  |
| Glycine                                  | 12.86                            | 8.7        | 11.3 |
| Alanine                                  | 13.61                            | 5.58       | 4.9  |
| Valine                                   | 17.83                            | -          | 1.6  |
| Methionine                               | 20.61                            | -          | 0.3  |
| Isoleucine                               | 22.72                            | 1.2        | 2.3  |
| Leucine                                  | 23.58                            | -          | 3.6  |
| Tyrosine                                 | 26.69                            | -          | 0.98 |
| Phenylalanine                            | 27.97                            | 5.4        | 1.3  |
| Histidine                                | 35.11                            | 3.1        | 3.2  |
| Lysine                                   | 39.16                            | 29.0       | 6.3  |
| Ammonia                                  | 39.16                            | 29.0       | 28.0 |
| Arginine                                 | 44.50                            | 1.1        | 3.1  |

Samples gave the same R<sub>t</sub> as standard  $\pm 0.2\%$ .

amino acids, such as phenylalanine and tyrosine, were not detected in sample 3H and were present in small amounts in 4H. Both samples appeared to be dominated by the higher percentage of ammonia, which may be attributed to the decomposition of amino nitrogen by the yeast deaminase (Samuel and Cecil 1959). The presence of considerable amounts of threonine gives an indication of the presence of a wild type of *Saccharomyces cerevisiae* which lacks the enzyme threonine deaminase, and cannot utilize it (Brunner et al. 1969). On the other hand, the presence of a high percentage of glutamic acid may be attributed to its synthesis from  $\alpha$ -ketoglutarate by yeast (Witt et al. 1964). Some amino acids such as valine, isoleucine, tyrosine and phenylalanine were possibly converted to the corresponding alcohol with the passing of time (Samuel and Cecil 1959, Fig. 2).

Although charring of the food has probably caused severe denaturation of the original peptides, the free amino acids detected are of neutral types which are characterized by a less active dipolar-nonvolatile nature. This is perhaps the reason for their survival (Cram and Hammond 1964).

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